

Aaron Schulz

Software Engineer

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SUMMARY

A strategic software engineer with 8+ years of transformative experience, I've architected solutions that slash infrastructure costs by 35%, reduce payment processing failures by 78%, and drive digital transformation across industries from finance to global retail. My technical arsenal spans **Azure** cloud ecosystems, **Angular** frontends, **.NET Core** backends, and **DevOps** methodologies, enabling me to craft high-performance, scalable technologies that solve complex business challenges. Passionate about turning technological potential into measurable impact, I excel at designing intuitive systems that not only meet but exceed organizational efficiency and user experience expectations.

EDUCATION

BAYLOR UNIVERSITY

Waco, Texas

Master of Science in Computer Science

May 2014 - Apr 2016

*** Key Focus Areas:** Advanced Algorithms, Database Systems, Software Engineering Methodologies, Operating Systems, Computer Networks, Research in Software Systems Development

Bachelor of Science in Computer Science

Apr 2010 - May 2014

***Key Coursework:** Software Engineering, Software Quality Assurance, Software Project Management, Engineering Economics, Data Structures and Algorithms, Operating Systems, Programming Languages (C++, C#, Java, Python, Javascript)

PROFESSIONAL EXPERIENCE

MERCURY INC

Houston, TX

Senior Software Engineer

Sep 2022-Present

Project Overview: Led the development of fraud detection and dispute resolution platforms, transforming complex risk management processes into intuitive, scalable digital solutions that enhanced customer experience and operational efficiency.

Key Achievements

- Architected and implemented a cloud-native microservices system on Azure, reducing infrastructure costs by 35% and improving system scalability
- Developed robust backend APIs using .NET Core and MS SQL Server, creating a high-performance database solution for fraud claim processing
- Designed and deployed a Angular frontend with NgRX state management, improving system performance and reducing user interaction time by 45%
- Implemented advanced Azure Cloud security protocols, enhancing data protection and compliance for sensitive financial information

Technical Contributions

- Engineered end-to-end authentication and authorization mechanisms using Azure Active Directory
- Developed comprehensive data migration strategies leveraging Azure Database services
- Created sophisticated data analytics dashboards using Azure Power BI for risk operations team
- Implemented robust error handling and logging strategies across cloud-native infrastructure

Skills Utilized

- **Frontend:** Angular 13/15, TypeScript, NgRx, Angular Material, Angular Universal, Angular CLI
- **Development Tools:** Nx, Docker, Webpack
- **Backend:** .NET 6/7, ASP.NET Core, C# 10, Entity Framework Core 6, Minimal APIs, Blazor Server
- **Cloud/Infrastructure:** AKS, Azure Service Fabric, Azure Container Registry, Azure Event Grid/Event Hubs
- **Security:** JWT authentication, ASP.NET Core Identity
- **Testing:** xUnit, NUnit, Moq, Jasmine, Karma, Cypress

WALMART

Bentonville, AR

Software Engineer

Aug 2019-Jun 2022

Project Overview: Developed a feature-rich, multilingual eCommerce platform supporting international operations across five countries, delivering a seamless shopping experience with advanced technological solutions.

Key Achievements

- Architected next-generation checkout solution operating across five countries, reducing international payment processing failures by 78%
- Championed comprehensive testing overhaul, implementing automated CI/CD pipelines that proactively identified and mitigated critical security vulnerabilities pre-production
- Translated complex business requirements into precise technical roadmaps with quantifiable performance metrics
- Rapidly resolved critical system outage affecting Mexican stores, implementing emergency fix within 4 hours, minimizing revenue impact

Technical Contributions

- Implemented advanced PayPal integration and sophisticated shopping cart system using Angular services and NgRx state management
- Created dynamic product catalog pages leveraging Angular's virtual scrolling for efficient rendering of large product collections
- Developed interactive product detail pages with comprehensive Angular Image Gallery for enhanced fashion item presentation
- Engineered robust internationalization support with ngx-translate and .NET Resource files, enabling seamless English and German user experiences
- Implemented comprehensive client-side and server-side validation strategies using Angular Forms and FluentValidation

Skills Utilized

- **Backend:** .NET Core 2.0-3.1, C# 7-8, ASP.NET Core MVC, Entity Framework Core, REST APIs
- **Frontend:** Angular 2-8, TypeScript, Angular Material, Bootstrap 4, NgRx
- **Cloud/Infrastructure:** Azure App Service, Azure SQL Database, Azure Storage, Azure CDN
- **Payment Processing:** PayPal REST API
- **Internationalization:** ngx-translate, .NET Resource files
- **Testing:** xUnit, MSTest, Jasmine, Karma
- **SEO Tools:** Angular Universal, Angular Meta

SIEMENS HEALTHINEERS

Software Developer

Walpole, MA

Feb 2017-Jun 2019

Project Overview: Pioneered innovative digital health technologies, developing cutting-edge web applications for VR therapy and medical device software that transformed patient care and medical technology outcomes.

Key Achievements

- Developed critical components for FDA-validated cardiac monitoring software, ensuring rigorous medical device compliance and patient safety
- Created advanced algorithms for diabetes management system, reducing false diagnostic readings by 45%
- Designed and implemented comprehensive cardiac and diabetes management platforms meeting stringent healthcare technology standards
- Optimized medical device software, dramatically improving battery efficiency by 30%
- Rapidly resolved critical production issues within 48 hours, minimizing potential patient care disruptions

Technical Contributions

- Built secure, HIPAA-compliant web interfaces using AngularJS that maintained modern UX standards while protecting sensitive health information
- Implemented robust authentication and role-based access control systems for protected health information management
- Created sophisticated audit logging mechanisms to track all PHI access and modifications
- Developed end-to-end encrypted data transmission protocols for secure patient communications
- Designed interactive data visualizations using D3.js to comprehensively display patient progress and therapy outcomes
- Collaborated extensively with QA teams to implement rigorous unit testing protocols

Tech Skills Utilized

- **Frontend:** AngularJS, JavaScript, Bootstrap 2-3, D3.js
- **Backend:** .NET Framework 4.0-4.5, C# 5, ASP.NET MVC 4-5, Web API
- **Medical Compliance:** FDA Validation, HIPAA Standards, HL7 FHIR

- **Cloud/Infrastructure:** Azure Cloud Services, SQL Azure
- **Security:** OAuth 1.0, WS-Federation, HTTPS/TLS
- **Testing Frameworks:** MSTest, NUnit, Moq, Karma, Jasmine
- **Communication Protocols:** RESTful APIs, JSON, AJAX, WCF

TOYOTA
Software Developer

Toyota City, Aichi, Japan - Plano, TX
May 2016-Jun 2017

Project Overview

Transformed enterprise infrastructure and development processes, driving efficiency, reducing operational overhead, and accelerating system delivery across complex technology ecosystems.

Key Achievements

- Built comprehensive developer portal consolidating five disparate systems, dramatically reducing engineering onboarding time from two weeks to just three days
- Implemented advanced CI/CD pipeline strategies that compressed deployment cycles from days to a mere 45 minutes
- Created innovative self-service infrastructure provisioning tools, slashing infrastructure request wait times from days to minutes
- Piloted strategic containerization initiatives that significantly reduced infrastructure and operational costs
- Developed sophisticated monitoring solutions that improved mean time to detection for critical system issues

Technical Contributions

- Designed and implemented cloud-native infrastructure using Azure Cloud Services
- Engineered automated deployment pipelines with comprehensive monitoring and logging
- Developed infrastructure-as-code solutions using containerization technologies
- Created scalable, secure microservices architecture supporting complex application ecosystems
- Implemented robust security and compliance frameworks

Tech Skills Utilized

- **Cloud Platforms:** AKS, Azure Functions, Virtual Networks, Load Balancers, Azure DNS, Azure Key Vault, Azure Pipelines, Azure Repos, CosmosDB
- **Containerization:** Docker, Kubernetes
- **CI/CD Tools:** Azure DevOps
- **Infrastructure as Code:** ARM Templates, Terraform
- **Monitoring:** Prometheus, Grafana, Azure Monitor
- **Backend Technologies:** .NET Core, Java, Python
- **Scripting:** PowerShell, Azure CLI, Bash, Python
- **Configuration Management:** Ansible, Chef
- **Virtualization:** Azure Virtual Machines, Hyper-V
- **Security:** OAuth, HTTPS/TLS, Azure Security Center